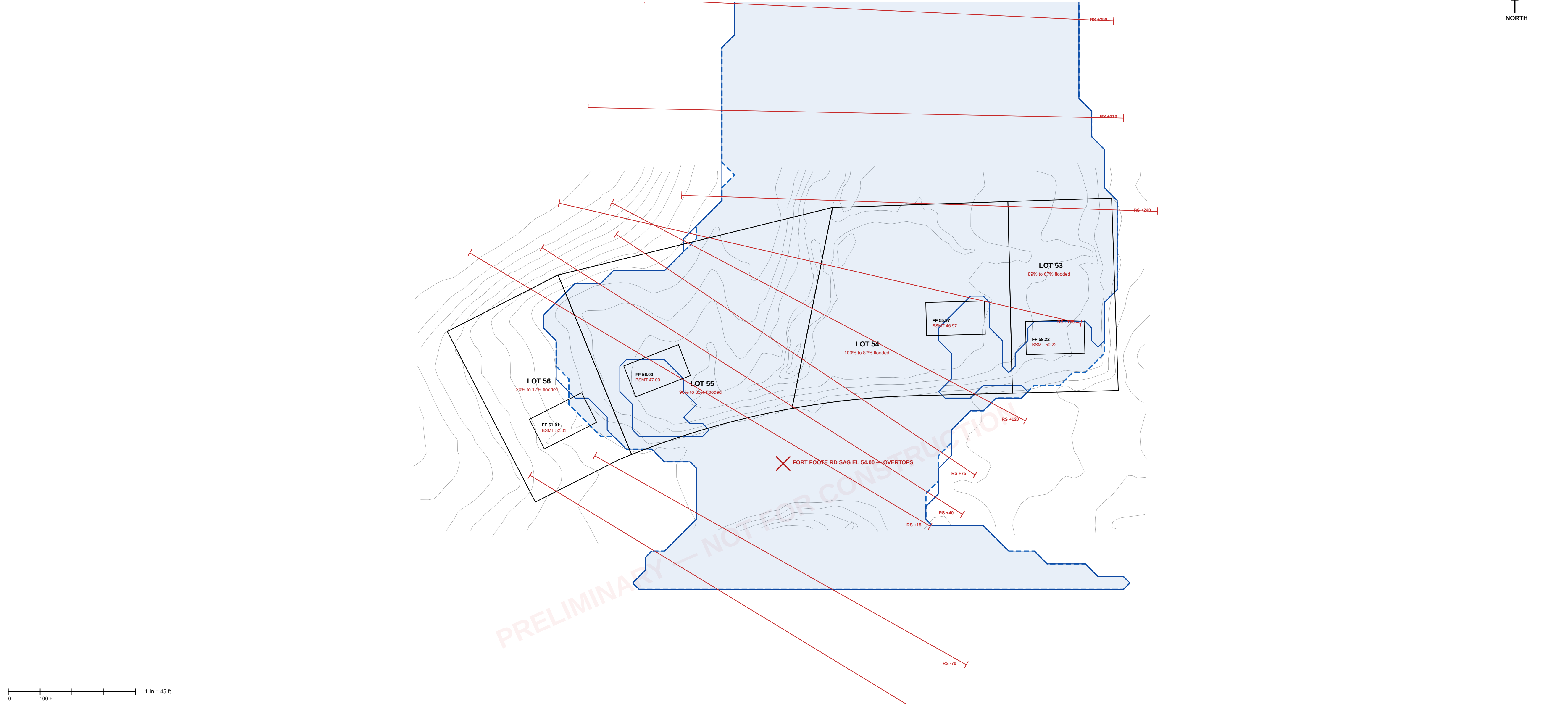
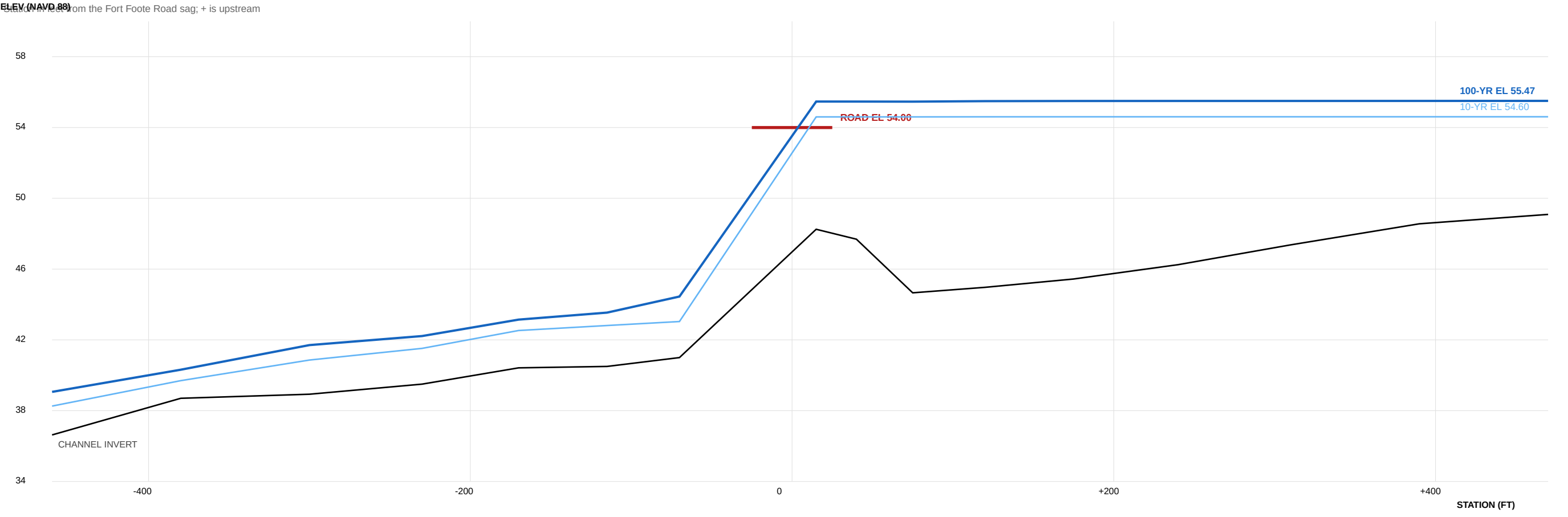


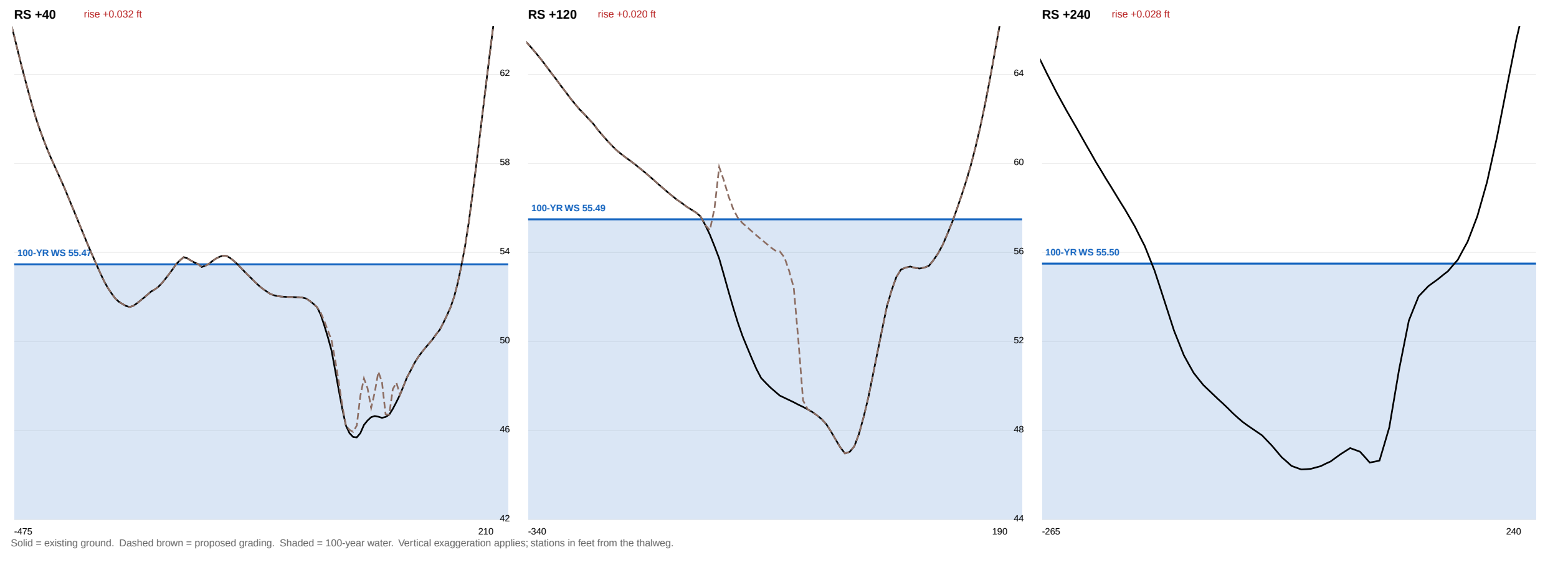
FLOODPLAIN DELINEATION — EXISTING AND PROPOSED, 100-YEAR



WATER-SURFACE PROFILE ALONG NORTH BRANCH BROAD CREEK



MODELLED CROSS SECTIONS — EXISTING AND PROPOSED



DESIGN DISCHARGES — NRCSTR-55, TYPE II					
STORM	P24 (IN)	Q (IN)	qw	PEAK (CFS)	
10-year	4.81	1.75	324	351	
25-year	6.01	2.63	350	571	
50-year	7.08	3.47	367	792	
100-year	8.29	4.48	382	1,061	
Area	397 ac	CN	68.0	Tc 0.86 hr	
CULVERT AT Q100					
36 in	inlet	HW 55.45	HWWD 3.6	939 over rd	
48 in	inlet	HW 55.36	HWWD 2.7	855 over rd	
60 in	inlet	HW 55.26	HWWD 2.1	757 over rd	
72 in	inlet	HW 55.13	HWWD 1.7	650 over rd	
84 in	inlet	HW 55.00	HWWD 1.5	541 over rd	
96 in	inlet	HW 54.87	HWWD 1.3	439 over rd	

EXISTING vs PROPOSED WATER SURFACE — 100-YEAR					
SECTION	EXISTING	PROPOSED	RISE (FT)		
RS +15	55.47	55.47	+0.000		
RS +40	55.47	55.50	+0.032	RISE	
RS +175	55.46	55.50	+0.039	RISE	
RS +120	55.49	55.51	+0.020	RISE	
RS +175	55.50	55.52	+0.026	RISE	
RS +240	55.50	55.53	+0.028	RISE	
RS +310	55.50	55.53	+0.029	RISE	
RS +390	55.50	55.53	+0.028	RISE	
RS +470	55.51	55.53	+0.028	RISE	
MAXIMUM			+0.039	NOT NO-RISE	
Storage removed below the 100-yr surface			2,908 cy		
Total fill in the modelled reach			3,666 cy		
Compensatory storage			REQUIRED		

FLOODPLAIN ON THE LOTS, AND FREEBOARD — 100-YEAR				
LOT	AREA (SF)	FLOODED EXIST	FLOODED PROP	FILL < FLOOD
LOT 53	12,500	11,100 (89%)	8,400 (67%)	281 cy
LOT 54	23,500	23,500 (100%)	20,500 (87%)	1,491 cy
LOT 55	25,600	24,500 (96%)	21,100 (82%)	1,038 cy
LOT 56	13,900	2,800 (20%)	2,400 (17%)	12 cy
LOWEST FLOOR CHECK AGAINST WS EL 56.53				
LOT	FIN FLOOR	FREEBOARD	BASEMENT FL	BELOW FLOOD
LOT 53	59.22	+3.69	50.22	5.31 ft
LOT 54	55.97	+0.44 FAIL	46.97	8.56 ft
LOT 55	56.00	+0.47 FAIL	47.00	8.53 ft
LOT 56	61.01	+5.48	52.01	3.52 ft

BASIS, METHOD AND LIMITATIONS

HYDROLOGY NRCSTR-55 graphical peak discharge, Type II distribution. Drainage area delineated by D8 routing on the USGS 3DEP bare-earth DEM. Curve number composed from the county 2023 impervious surface and tree canopy layers and the NRCS soil survey. Rainfall from NOAA Atlas 14 Vol. 2 Ver. 3, partial duration series, at 38.7611 N 77.0115 W.

HYDRAULICS One-dimensional steady gradually-varied flow, standard step, conveyance subdivided at the bank stations, average conveyance friction slope. Crossing analyzed to FHWA HDS-5 with both controls computed and the roadway treated as a broad-crested weir.

TERRAIN POKas Contour 2 ft (2023), NAVD 88, interpolated to a 10 ft grid. THIS IS NOT A FIELD SURVEY.

DATA NAVD 88 + NGVD 29 - 0.86 ft here, from NGS marks HV4762, HV4763 and HV4728. On that basis the 57 ft WSSC of FPS-770017 is about EL 56.3 NAVD 88, against EL 55.47 computed. To be confirmed by leveling to benchmarks H31A, H32A and H32B on the field topographic survey.

NOT DONE No field survey. The existing culvert has never been measured and its size, inverts and entrance are assumed. HEC-RAS was not run and no HEC-RAS files exist. FPS 200546, the controlling study of record, has not been obtained. Upstream stormwater facilities were not inventoried. No floodway was delineated and none exists on current mapping. No unsteady or storage routing.

FLOODPLAIN OF RECORD FEMA FIRM panel 2403C02202E eff. 2016-09-16 maps these lots Zone X, outside the special flood hazard area, with no BFE and no floodway. That is a statement about FEMA mapping, not about flood risk. The county regulates this floodplain through the FPS series regardless.

EASEMENT The only patented floodplain easement in Indian Queen East (Pat 118-083, recorded 17 January 1984, 5.32 ac) lies entirely south of Fort Foote Road and covers none of Lots 53-56.

KEALEE

Design, build, deliver

PROJECT
Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS
9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION
Prince George's County, Maryland

ZONE
RSF-95

PLAT OF RECORD
Plat Book WWW 65, folio 60

SHEET
FP-101 — FLOODPLAIN STUDY AND DELINEATION, EXISTING AND PROPOSED

DISCIPLINE
Maryland Professional Engineer

HORIZONTAL DATUM
NAD 83, Maryland State Plane, US survey feet (EPSG:2248)

VERTICAL DATUM
NAVD 88

DATE
2026-09-11

STATUS
PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. No field survey. Culvert never measured. HEC-RAS not run. FPS 200546, the controlling study of record, not obtained. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. _____ Date _____

KEY FINDINGS

Contributing drainage area 397 ac (0.620 sq mi)
Composite curve number 68.0
Time of concentration 0.86 hr
Q100 at the crossing 1,061 cfs
Fort Foote Road sag EL 54.00
100-yr water surface, existing EL 55.47
100-yr water surface, proposed EL 55.53
Max rise, proposed vs existing +0.04 ft — NOT no-rise
Flood storage removed by fill 2,908 cy
Road overtops at Q100 956 cfs over the pavement

LEGEND

— 100-yr floodplain limit, EXISTING
— 100-yr floodplain limit, PROPOSED
— Modelled cross section
— Lot boundary of record
— Existing contour (2 ft, county LIDAR)